Design and Implementation of a Chorded On-Screen Keyboard for People with Physical Impairments

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Abstract. The purposes of this study were to design an alternative on-screen keyboard for people with physical impairments and to evaluate the efficacy of the chorded input method. The approach of the on-screen keyboard is based on the human computer interface. It gives visual guide and instant feedback to show users where they can find the characters they need. The system has been designed with the principles of universal design. Three factors including the learning ability, efficiency of using and subjective satisfaction are considered as the usability evaluation. According to a preliminary study, the participant felt mastered the text input method quickly. An experimental evaluation on the typing performance of the subjects with muscular dystrophy will be measured under both scanning input mode and chorded input mode in the future.

1 Introduction

Computer technology has become an integral part in our daily life. It also helps people with physical impairments expand opportunities for educational success and improve their opportunities in both vocational settings and leisure activities. Text entry is widely regarded as one of the most frequent human-computer interactions. Using standard keyboard may seem pretty straightforward to most people. But for people with postural limitations, muscle weakness or limited movements, applying computer access are difficult. They encounter difficulties in various aspects when using standard keyboard, resulting in high error rates, fatigue and inefficiency. On-screen keyboards provide an image of a standard or modified keyboard on the computer screen by allowing users to select keys with a mouse, touch screen, trackball, joystick, or switch. Although the QWERTY layout is entrenched for standard keyboards and most of the on-screen keyboards are designed based on QWERTY layout. It is not a good user interface for people with physical impairments. The efficiency of the text entry depends on the suitability of control interface, which includes proper selection methods

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and keyboard layouts [1]. People who can manipulate only one hand, or people with more severe physical difficulties, may find the distance they have to travel to reach the whole keyboard restrictive. Such people may find that a more compact keyboard layout makes typing more efficient and less tiring for them. The purpose of this study was to design and implement an innovative on-screen keyboard for people with physical impairments which based on human computer interface. In the next sections the design of the on-screen keyboard and a pilot study will be discussed.

2 Design the On-Screen Keyboard

2.1 The Numeric-Based Input Method

Chorded keyboards use a combination of a few keys to create keystrokes for each letter. One clear problem is that typing text with the chorded keyboard is slower than using the QWERTY keyboard for novice user. The primary cost of the chorded keyboard is the extensive learning required associating the finger combinations with their corresponding actions, but after extensive practice, chorded keyboards have been found to support more rapid word transcription processing than the QWERTY keyboard, possibly due to reduced movement-time requirements [4]. Chorded keyboards are useful for people with limited arm range. Our chorded keyboard design efforts focus on input speed and learning time for people with physical impairments. It allows user to enter data and commands with one hand. We applied the numeric-based input method works by the user presses two numeric keys to generate character [3]. The interface design makes it simple and intuitive for users. To help users with their writing and editing, the letters will be pronounced when typed. It gives feedback to the user both visually through highlighting and through audio. As shown in Figure1, the first key press selects a group, and the second key select a letter, symbol or command in that group. For example, in order to type the letter ‘a’, the user first presses ‘4’ on the input device as shown to the left in Figure 1. The system will produce visual feedback to the user by highlighting the group of letters. And then, the user knows what group is activated. When the user presses ‘7’ (the second key), which selects the corresponding letter ‘a’ in the group. If a typing error was found, the user could use ‘0’ as a cancel button any time. For movement efficiency based on Fitts' law, the symbols and commands were configured by their frequency to reduce the fingers' traveling distance. For example, space is the most frequently used key and it is located on the center of the layout to reduce physical effort. The period mark is the most frequently punctuation, its corresponding two numeric keys are the same (2, 2). The travel distance of comma, question mark, and exclamation mark are shorter than other less frequently used punctuation mark. Although the frequency layout is theoretically physically fastest, for novice users, the entry speed is determined mostly by the needs to search and find target keys rather than by the amount of motor movement. The keyboard layout with alphabetical ordering will be faster initially for novice users.

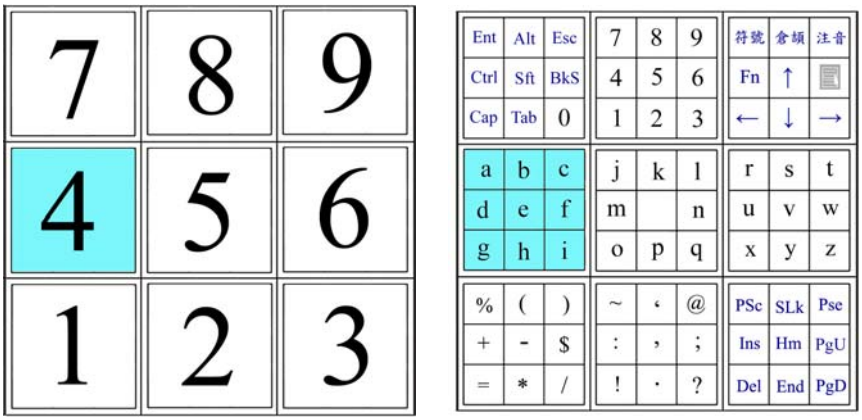

Fig. 1. Numeric-based Input method support visual feedback

2.2 Keyboard Layouts

Our chorded keyboard has been designed following the principle of universal design. It is displayed within a window that user can move, resize the layout and gives feedback to the user through audio. The system includes 9 layouts (international alphabetic layout, Scaffolding layout, Internet layout, 2 types of symbol layout, transparency layout, high contract layout and 2 types of Chinese Input layouts). The system can work with multiple access devices, such as keyboard, joystick, touch screen, touchpad, mouse, adapted switch. It also supports five alternative input methods (numeric-based mode, point-and-click mode, dwell mode, group-row-column scanning mode and joystick select mode). The users also could create their individual virtual keyboards by rearrange the special layout. Figure 2 outlines some keyboard layouts we implemented.

The scaffolding layout does not require much practice, and then can help novice user to reach a reasonable typing speed. The hint is superimposed directly on the target letter and the learner could pay attention to the same point. We suggest the novice user use the scaffolding layout at the first time. For visual impaired users, using the high contract layout, enlarging the size of the layout and using the option to pronounce letter may make a significant difference in the efficiency of their text entry task. Internet services also have particular benefits and potentialities for people with physical impairments. One of the useful applications for user is the World Wide Web (web). The keys on our internet layout are configured into three sets. One set for fast keys (1 in the Fig. 3), another for the command keys of browser (2 in the Fig. 3) and the last part is the same with the international alphabetic layout. Function key layout organized the common symbols, function keys and the common commands of the word processor (Microsoft word). The system includes a pop-up menu and allows reconfiguration as show in Figure 4.

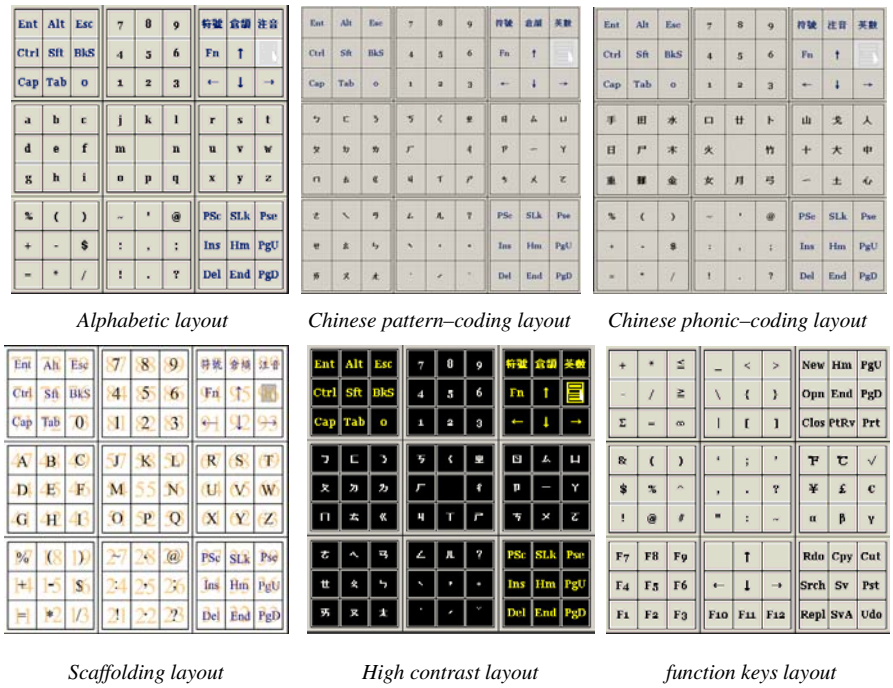

Fig. 2. The chorded keyboard layouts

Fig. 3. The internet layouts

Fig. 4. Pop-up menu

2.3 Group-Row-Column Scanning

Developing scanning mode assistive systems for people with physical impairments is not a new issue. The scanning approach is designed for people who fail to move the mouse pointer with any pointing device. Single switch automatic row-column scanning is the most common scanning approach for the people with severely physical impairments [5]. Beyond row-column scanning, we created group-row-

column scanning based on our square layout. We predict that it is more efficient than row-column scanning with one switch activation. Figure 5 shows the display associated with the group-row-column variety of scanning. First, each group is automatically scanned in sequence. When the highlight arrives at the desired group, the user hits the switch (figure 5(a)). The highlight then advances scans each row in that group until the user hits the switch again (figure 5(b)). Finally, the highlight scans across each item in the selecting row until the user hits the switch (figure 5(c)). Scanning input method is less cognitive effort when compared to the chorded input method.

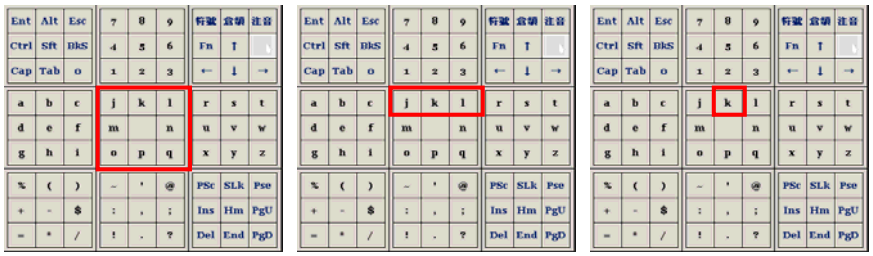

Fig. 5. Group-row-column scanning: the on-screen keyboard for single switch input

3 Usability Evaluation

To evaluate the usability, text entry tasks were performed to measure the typing performance of participant. Our initial trials for usability and efficiency have been restricted to one able-bodied college student. She was fluent in reading English and familiar with standard keyboard layout. The experiment started with an experimenter explaining the task and how to operate the chorded keyboard. And then, the subject took 15 minutes practice with the scaffolding layout and was asked to carry out typing test using a QWERTY keyboard and numeric-based input method. Each input method lasted 10 minutes. A picture of the test application is shown in Figure 6. An article is displayed on screen and the user is asked to type another copy and allowed to fix mistakes. Articles contain capital and lower-case letters, quotation marks and other punctuation. The test text for each trial was randomly selected from 42 articles, and both the typing speed and accuracy were recorded by the typing test program. Most people with physical impairments use only one hand for text entry, and novice user always use one finger for text entry task. In order to simulate the typing performance of single-digit user, the subject was asked to use a pencil instead of single finger. The subject must to carry out two keyboard patterns one trial a day, five days a week. We obtained ten measurements of the text entry speed as show in Figure 7. The performance of chorded-based, subject could type only 23.9 correct characters per minute in the first trial, and eventually up to 64.7 characters per minute in the last trial.

Fig. 6. Typing performance test

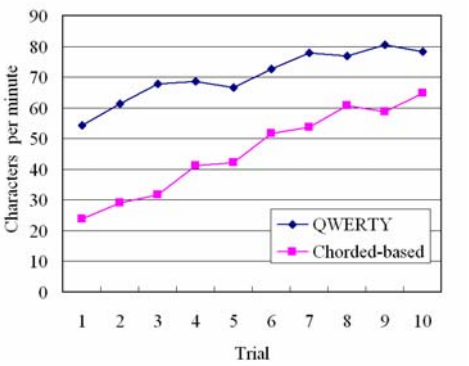

Fig. 7. Comparison of typing speed for QWERTY and chorded keyboard

Although the subjects' keystroke speed of chorded keyboard was slower than QWERTY keyboard, this might due to the subject's familiarity with QWERTY keyboard layout. However, the learning curve of chorded-based input method is very short. We also wanted to know how subject feel about the chorded input concept through post experiment interview. The interview was conducted after the final session. Subject was also encouraged to speak out and to write down comments or suggestions. The subject indicated that the numeric-based input method is easy to understand and the scaffolding layout is helpful for a novice learner. After using chorded keyboard for half of the third session, the subject did not even notice the concept any longer.

A potential limitation of this study is its reliance on a purely able-bodied subject, which may limit generalization the research results to the persons with physical impairments. However, we are confident that practice will give significant performance improvements and the chorded keyboard is more useful for some people with physical impairments.

4 Multiple Accesses

In order to make our chorded keyboard more useful for people with physical impairments, we have continually evolved the chorded input concept for user to multiple accesses. The system allows us to combine a wide variety of physical devices and language sets to form an interface ideally suited to the specific abilities of the user. The system can combine to combine other physical devices, such as N-fingers input device [2], USB calculator or PDA (Personal Digital Assistant) for some persons with muscular dystrophy (see Figure8 and Figure9). For persons with cerebral palsy, combine defined switches as input devices will help them computer access easily.

Fig. 8. USB calculator input device

Fig. 9. PDA input device

5 Conclusions

Although a large number of assistive technology products are available today, many people with physical impairments do not have adequate input devices. Most people with severely physical impairments have never been involved in computer usage throughout their life, interface designers must make the most of the metaphors involved. The need for participation in an emerging information society has let to several research efforts for designing accessibility solutions for people with physical impairments. Text entry for people with a motor impairment is a painfully tedious task. In this paper we present an alternative on-screen keyboard for them. We hope that the on-screen keyboard will contribute to creating accessible public computing facilities widely that can be enjoyed by people with physical impairments. Over the next few months, we will be improving the prototype and will then be offering it for people with cerebral palsy or muscular dystrophy.

We just begun our study of how the chorded keyboard can be used to help people with physical impairments, and there is much work to be pursued in many areas. We were able to put together several recommendations for the improvement of our on-screen keyboard based on the observations and the comments provided by the participant of the study. There are four implications and suggestions as following:

1. *Optimized keyboard layout.* The order of the letters in the scanning keyboards is crucial for their performance. To enhance the performance of our group-row-column scanning approach, the more frequently used keys could be rearranged together and to place the most frequently occurring characters in the most easily accessed locations.
2. *Word-prediction mechanisms.* Word-prediction is popular among devices that support text input. The on-screen keyboard embedded word-prediction mechanisms, characters prediction or abbreviation expansion will improve users' text entry speed.

3. *Embed intelligent system.* Reducing visual search for the selection will accelerate the typing speed. What's more, when an intelligent system that can recognize user's typing patterns is embedded, it will increase the performance.
4. *Voice recognition.* Voice recognition is an alternative input method. We will develop the voice command for our prototype in the future.

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